



XR-OOM: Mixing Virtual Driving Simulation with Real Cars and Environments Safely

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ABSTRACT

High-fidelity driving simulators can act as test beds for designing in-vehicle interfaces or validating the safety of novel driver assistance features. XR-OOM is a mixed-reality driving simulator system that enables us to superimpose virtual objects and events into the view of participants engaging in real-world driving in unmodified vehicles. In this work-in-progress paper, we present system requirements and specify our initial proof-of-concept system design. This work enables us to further develop measures for using such systems in ways that are safe, effective, and lead to accurate, repeatable data collection about behavioral responses in hybrid real-world/simulated driving tasks.

CCS CONCEPTS

• **Human-centered computing** → **HCI design and evaluation methods**; *Virtual reality*; *Interface design prototyping*; Usability testing.

KEYWORDS

XR, mixed reality, driving simulation, automotive, user studies, design, safety

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1 INTRODUCTION

Driving simulation can be used to test the usability, safety, and driver performance impacts of interfaces and interactions. Simulators can repeatedly recreate scenarios that are rare or risky in the real world,

offer experimental control over test events and conditions, and allow for rapid prototyping and testing. [5]

Fixed-base simulators in laboratories often feature graphical simulation environments, as well as screen-based instrument panels and center consoles to enable flexibility in interfaces and scenarios for experimental design. In comparison, actual on-road or test-track vehicles can better incorporate external factors intrinsic to driving, such as vibration and physical motion, but typically require far more effort to vary context, scenario design, critical event design and user interface for experiments.

We seek to combine the strengths of immersive simulation environments and real-world testing environments in a mixed reality on-road driving simulator (XR-OOM). Using a video pass-through mixed reality headset, we create simulated events and interfaces while drivers operate a real vehicle. In this work-in-progress paper, we present system requirements and a proof-of-concept system design. This is the first work in automotive user interaction literature to describe using mixed reality systems for real-world driving simulation. This work contributes a critical step towards realistic and safe development environments for human-centered design and testing of future vehicle interfaces.

2 DESIGN REQUIREMENTS

In the next section, we introduce system requirements that need to be met in order to achieve the required levels of safety, immersion and comfort, as well as to enable a range of experimental studies. These stem from our background research, as well as early stage experiments performed to understand how to safely, comfortably and reliably perform mixed-reality driving simulation in our passenger vehicle.

2.1 Experimental range

In order to support various experimental studies, the system should ideally include:

Environmental Objects Driving simulator studies often feature external objects, such as pedestrian or bicycle road incursions, road signs or visual distractions. The system needs to be able to simulate objects in the environment outside the car.

Environmental Phenomena To explore how environmental factors like weather, time of day, or extraordinary traffic conditions affect user experience and vehicle response, the system would need

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Figure 1: Early testing example of using the XR-headset inside a research vehicle in a restricted testing area. Top: Driver wearing headset. Bottom: View of headset pass-through video with virtual cone objects.

to be able to simulate environmental phenomena in the view the driver has outside the vehicle.

Display Technologies Since novel driving paradigms will require new displays and controls, the system should be possible to visually alter or change any display surfaces, including the instrument cluster and center console. Similarly, heads-up displays and other new display technologies should be able to be prototyped or evaluated in physical space easily. For example, the system environment may want to respond to external objects—highlighting pedestrians it senses and recognizes, for example—or feature simulated objects or actors to distract or in other ways impact the participant.

Device Technology Because brought-in devices such as phones or tablets are regularly used by drivers to assist with driving tasks, the system should also be able to support the incorporation of these devices and record participant use and response to these systems.

Visual Impairments A lot of driving simulation research focuses on people with various impairments, and on designing compensatory measures for such drivers. The system could simulate the impact of visual impairments like limited field-of-view on driving.

Locus of Control To support research into driving automation, the system could support driving automation, such as teleoperation [14] or supervisory control [10] scenarios and control schema.

Interactive Technologies Since interactive technologies critically affect driving by distracting or assisting drivers, it is important that the simulation system be able to support novel interaction scenarios by enabling the testing of social agents and voice interaction. Conversely, social agents could prevent drivers from becoming bored or angry in the vehicle, or help point out important external cues to improve situation awareness.

Behavioral Capture The behavioral response of drivers to the simulated scenarios and experimental task needs to be captured in ways that contextualize their response. This means that both the physical and virtual environment need to be captured with enough resolution to recreate situations for analysis.

2.2 Visual Resolution

The visual resolution influences the participant's level of immersion and their ability to perform physical actions in the vehicle. The required visual resolution affects not only the screens inside the head mounted display, but also the cameras used to capture the experimental environment.

One of the essential interaction with a vehicle interface is reading dials, icons, and texts, as such displays providing road directions and feedback on vehicle performance. Text readability has been a serious concern since the early days of virtual reality [8, 11]. Based on work by Dingler et al. [4], the resolution required is a function of the text size and rendered distance. Curevo et al. found that “life-like” virtual reality requires a resolution of 60 pixels per degree for 20/20 vision. [3]

2.3 Field of View

The field of view (FoV) requirement is a function of people's peripheral perceptions. Van Erp and Padmos argue that FoV is critical to lateral control tasks, leading to a stronger impact on driving performance than visual resolution or latency [19]. Driving performance was higher in drivers with a 100 degree field of view compared to those provided with a 50 degree FoV. Hu et al. note that the human eye has a $210^\circ \times 150^\circ$ (i.e., diagonal 200°) field of view, where existing commercial VR headsets only achieve a maximum 150° of FoV.

The field of view in the system should at least support spotting traffic, and visibility of all mirrors.

2.4 Pass-through and Rendering Response Latency

Research on the combined latency effects on driving performance have recently come from studies of remote teleoperation of driving. Neumeier et al. [15] determined that even a delay of 300ms will cause a deterioration of the driving performance. Liu et al. [12], identified network delay (LTE) variability is a stronger predictor of negative remote driving performance than the maximum delay magnitude. They found that the maximum tolerable latency is 358ms.

In studies of networked multi-player racing games, the acceptable delay is up to 50ms [2, 16].

Virtual Reality headsets also pose requirements on visual delay, as shown by Curevo et al. [3]. Mania et al. [13] identified the Just Noticeable Difference point to be at about 15ms with up to 20ms

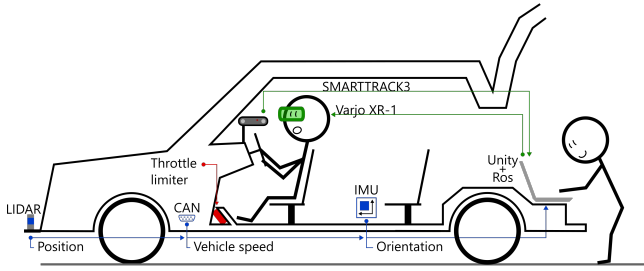


Figure 2: This diagram shows the main layout of the sensors for the vehicle (LIDAR, IMU, ODOM), the internal tracking (SMARTTRACK3) and rendering with a laptop and XR headset.

being referenced as the maximum before performance and immersion starts to be impacted. Similarly, Hu et al. indicate that less than 20ms latency is recommended to avoid motion sickness and discomfort [9].

2.5 Participant Comfort

With mixed reality in a moving vehicle, both motion- and simulator sickness need to be avoided. Counterintuitively, both Paredes, et al. and Goedicke, et al. found that, with a high resolution virtual reality headset in a vehicle with well-correlated physical and virtual motion, the combination of the virtual reality environment and the physical environment causes less motion sickness than virtual reality environments or in-vehicle experiments themselves [7, 17].

Furthermore, heavy or unbalanced headsets can create discomfort in participants over time. Hence, weight limits how long participants are comfortable to do an experiment. Early systems, like the Bachelder’s Fused Reality system, needed a counterweight system to compensate this issue [1].

3 SYSTEM DESIGN

Our proof of concept XR-OOM system intends to meet the aforementioned requirements of a mixed reality in-vehicle driving simulation environment. We anticipate that the viability of such a system will increase as technologies improve, and that our planned validation of the approach can inform possibilities and development for this research platform. In the current configuration, the systems operates at an almost constant 60 - 90 Frames/s and with a frame delay between 35 and 25ms. The car relative tracking with the ZED2 operated at 60 frames/s while the headtracking operates at up to 240 Frames/s with a 10 ms delay.

3.1 Platform Selection

While our system is based on the Varjo XR-1 which already demonstrated the potential to be used in a vehicle¹ and later published by Ghiurău et al. [6]. Similar to their setup we made a few additional platform decisions beyond the headset to enable an more complete experimental platform:

¹<https://vimeo.com/339100197>

Vehicle Toyota Prius Alpha is used as a base vehicle without modification except the installation of the devices described below. The vehicle has an on-board diagnostics port that allow us to read vehicle speed from CAN (Controller Area Network) bus. A throttle limiter is installed to keep the vehicle speed within a range that allows the experiment to be performed safely.

Headset Varjo XR-1² and XR-3 are currently the only headsets meeting most of our fidelity requirements. The visual delay between the cameras to headset screen is between 14ms to 30ms, meeting the requirements for remote vehicle operation and immersive VR.

Rendering Engine This engine renders virtual objects that are superimposed on the participant’s view. The most prevalent choices are the Unreal³, and Unity⁴ engines. The current version of XR-OOM uses Unity (2020.4 LTS). However, Unreal Engine could be a viable alternative.

Rendering Computer The computational requirements are driven by requirements from the Varjo XR-1 headset. The rendering computer also handles gaze tracking. Extensive work went into finding a laptop that could support this setup. However, thermal, and power limits caused issues even if the hardware specs (i.e. Version numbers) matched the requirements. We finally built a custom computer with a 9900K i7 CPU Nvidia RTX 2080Ti and a motherboard supporting thunderbolt. To power the computer in the vehicle, we connected three deep-cycle 12V Batteries in parallel to a 1.2kW modified-sine-wave converter. This setup powers the computer long enough to run about 2 experiments each of which go for about 40 minutes before the batteries need to be recharged.

Head Tracking The participant head pose needs to be tracked relative to the vehicle cabin to accurately render the scene from the driver’s perspective. Head tracking is achieved with The SMARTTRACK3 from ART⁵. With custom tracking solutions, calibration needs to be done manually, where accurate horizon calibration (roll and pitch) is paramount.

Vehicle Localization Localizing the pose of the vehicle in the real world is essential. In combination with tracking the headset relative to the vehicle, we can place virtual objects in to the real world from the perspective of the participant.

We are using localization libraries running inside open-source Robot Operating System (ROS)[18], in combination with a Lidar (Oyster OS-1), IMU (Xsens MTI-300), and CAN-Bus Speed readouts from the vehicle to localize the vehicle in the real world. We use a ZED-2 for additional localizing and mapping work directly inside Unity. Should one of the two system (ROS or Zed-2) loose tracking, one serves as a back-up.

4 CONCLUSION

In this report, we introduced our novel on-road mixed reality driving simulator and described its technical details. We are currently developing procedures for an evaluative user study. To that end, we devised a driving task (see Figure: 4 around traffic cones (see Figure: 3). Using quantitative measures including task time, average

²<https://varjo.com/products/xr-1/>

³<https://unrealengine.com>

⁴<https://unity3d.com>

⁵<https://ar-tracking.com/products/tracking-systems/smarttrack3/>



Figure 3: XR-OOM can run standalone in a functional vehicle. Shown above is the XR-OOM setup being used in a driving study.

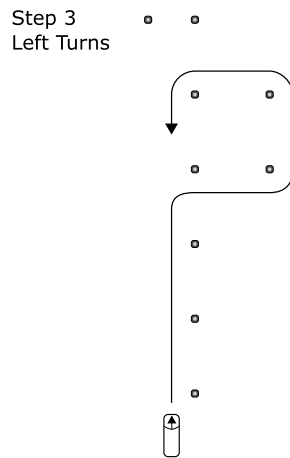


Figure 4: Example of one of the driving tasks to be performed both with and without the headset.

driving speed, and turning radius as well as qualitative analysis including questionnaire and video coding, and post interviews, we hope to evaluate how the participants' driving performance is impacted by the headset, and to what extent their reactions are different or similar between real and virtual objects.

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